

**2019 Mathematics Kangaroo Contest****—Cadet—****2019 数学袋鼠竞赛****Part 1: 10 problems, 3 points each | 第一部分: 10 道题目, 每题 3 分**

1.  $20 \times 19 + 20 + 19 =$

- (A) 389 (B) 399 (C) 409 (D) 419 (E) 429

2 • A model train takes exactly 1 minute and 11 seconds for each round on a course. How long does it take for six rounds?

火车模型在一条轨道上转一圈需要 1 分 11 秒。那么转六圈需要多少时间?

- (A) 6 minutes 56 seconds 6 分 56 秒 (B) 7 minutes 6 seconds 7 分 6 秒
- 
- (C) 7 minutes 16 seconds 7 分 16 秒 (D) 7 minutes 26 seconds 7 分 26 秒
- 
- (E) 7 minutes 36 seconds 7 分 36 秒

3 • A barber wants to write the word SHAVE on a board in such a way that a client looking into the mirror reads the word correctly. How should the barber write it on the board?

理发师想要在板上写一个单词 SHAVE, 使得客人从镜子里看就能正确的读出这个单词。问理发师应该如何写?

(A) SHAVE  
(D) EVAH2(B) SHAVE  
(E) EVAH2

(C) EVAHS

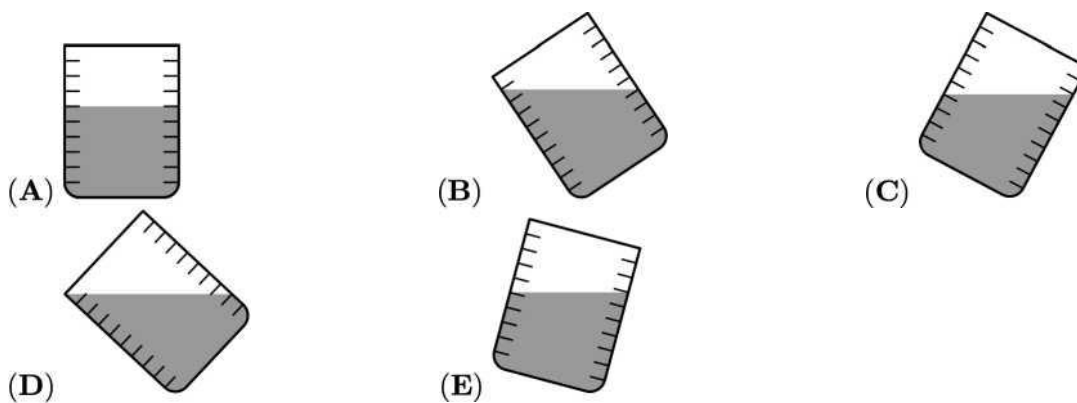
4 • How many different sums of dots can you get by rolling three standard dice simultaneously?

同时抛掷三个标准的骰子所得到的点数之和有多少种不同的可能?

- (A) 14 (B) 15 (C) 16 (D) 17 (E) 18

5 • Five identical glasses are filled with water. Four of them contain the same amount of water. Which one contains a different amount?

五个相同的玻璃杯里都装着水。其中四个杯子里的水一样多。问哪个杯子中的水与其他杯子里的水不一样?



6 • A park has five gates. Monica wants to enter through one gate and to exit through a different one. In how many ways can she enter and exit the park?

一个公园有五个门，Monica 想通过一个门进入并通过另一个门离开。问她有多少种方式进出公园？

- (A) 25                      (B) 20                      (C) 16                      (D) 15                      (E) 10

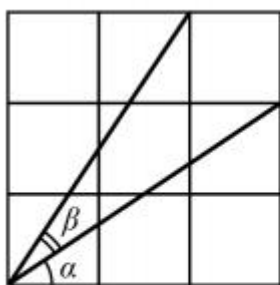
7 • The weight of each of three kangaroos is a different whole number in kilograms. The total weight of them is 97 kilograms. How many kilograms can the lightest of them weigh at most?

三只袋鼠的重量互不相同，都是整数千克。它们的总重量为 97 千克。问最轻的那只袋鼠最多可能是多少千克？

- (A) 1                      (B) 30                      (C) 31                      (D) 32                      (E) 33

8. Which of the following statements is true for the marked angles in the given figure of nine identical squares?

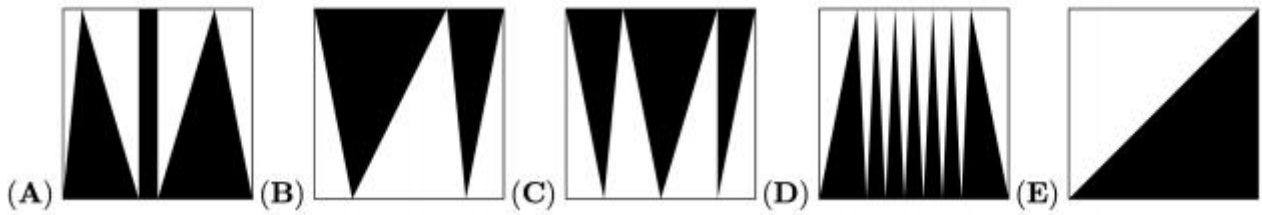
角度标记在给定的由九个正方形组成的图形中，下列哪一项论断是正确的？



- (A)  $\alpha = \beta$                       (B)  $2\alpha + \beta = 90^\circ$                       (C)  $\alpha + \beta = 60^\circ$   
 (D)  $2\beta + \alpha = 90^\circ$                       (E)  $\alpha + \beta = 45^\circ$

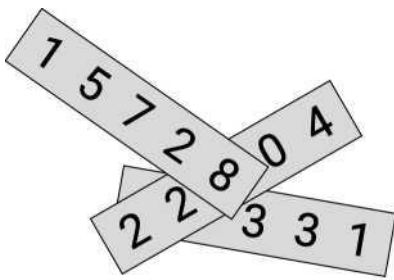
9. Inside each unit square a certain part has been shaded. In which square is the total shaded area the largest?

在每个单位正方形内部，都有一部分都涂成了黑色。哪个正方形中，黑色的总面积最大？



10 • On each of three pieces of paper a five digit number is written as shown. Three of the digits are covered. The sum of the three numbers is 57263. Which are the covered digits?

如图所示，在三张纸上分别写有五位整数。有三个数字被覆盖了。这三个数的总和是 57263。问被盖住的是哪个数字？



- (A) 0, 2, 2      (B) 1, 2, 9      (C) 2, 4, 9      (D) 2, 7, 8      (E) 5, 7, 8

**Part 2: 10 problems, 4 points each | 第二部分：10 道题目，每题 4 分**

11. A square has vertices A, B, C, D labelled clockwise. An equilateral triangle is constructed with labels A, E, C labelled clockwise. What is the size of angle CBE in degrees?

一个正方形的顶点按顺时针方向记为  $ABCD$ 。构造一个等边三角形，使得沿顺时针方向，它的三个顶点依次是  $A, E, C$ 。问角  $CBE$  的大小是多少度？

- (A) 30      (B) 45      (C) 135      (D) 145      (E) 150

12. The numbers  $a, b, c, d$  are distinct positive integers chosen from 1 to 10. What is the least possible value  $\frac{a}{b} + \frac{c}{d}$  could have?

数  $a, b, c, d$  是从 1 到 10 中选择的不同正整数。 $\frac{a}{b} + \frac{c}{d}$  可以取得的最小可能值是多少？

- (A)  $\frac{3}{19}$       (B)  $\frac{2}{10}$       (C)  $\frac{14}{45}$       (D)  $\frac{29}{90}$       (E)  $\frac{25}{72}$

13 • The flag of Kanguria is a rectangle with side lengths in the ratio 3 : 5. The flag is divided into four rectangles of equal area as shown. What is the ratio of the side lengths of the white rectangle?

Kanguria 的旗帜是一个边长比为 3 : 5 的矩形。如图所示，这个旗帜被分成了四个面积相等的矩形。问白色矩形的边长的比例是多少？



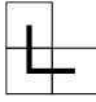
(A) 1 : 3

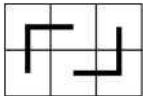
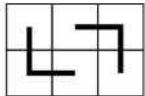
(B) 1 : 4

(C) 2 : 7

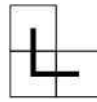
(D) 3 : 10

(E) 4 : 15

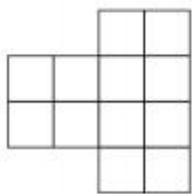
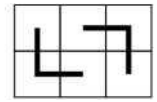
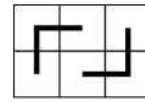
14. A  $3 \times 2$  rectangle can be exactly covered by two of the L-shape figures  in two different ways as

shown below :  . In how many different ways can the figure below be covered by the L-shape figures?

一个  $3 \times 2$  的矩形可以有两种不同的方式被两个 L 形图形恰好完全的覆盖。问下图有多少种不同的方式被 L 形图形覆盖？



恰好完全的覆盖。



(A) 1

(B) 2

(C) 3

(D) 4

(E) 48

15 • The triathlon consists of swimming, running, and biking. The biking is three-quarters of the total distance; the running is one-fifth; and the swimming is 2 kilometers. What is the total distance of this triathlon, in kilometers?

铁人三项包括游泳，跑步和自行车。自行车的赛程是总赛程的四分之三；跑步的赛程是五分之一；游泳的赛程是 2 千米。问铁人三项的总赛程是多少千米？

(A) 10

(B) 20

(C) 38

(D) 40

(E) 60

16. Some diluted juice is to be made out of concentrate and water in the ratio 1 : 7 by volume. Juice concentrate is in a 1-litre flask, and the flask is half full. What fraction of this concentrate should be used to produce 2 litres of diluted juice?

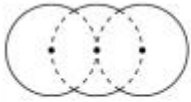
稀释果汁由浓缩果汁和水按 1 : 7 的体积比制成。浓缩果汁在一个 1 升的容器中，容器是半满的。应该使用这些浓缩果汁的几分之几来制作 2 升的稀释果汁？

(A)  $\frac{1}{4}$ (B)  $\frac{1}{2}$ (C)  $\frac{2}{7}$ (D)  $\frac{4}{7}$ 

(E) All of the concentrate. 全部浓缩果汁

17 • The given shape is made of parts of three equal circles of radius  $R$  that have their centres on a straight line. The middle circle passes through the centres of the other two, as shown. What is the perimeter of the shape?

给定的形状由三个相同的半径为  $R$  的圆组成，三个圆的圆心都在一条直线上。如图所示，中间的圆穿过另外两个圆的圆心。问这个形状的周长是多少？



(A)  $\frac{10\pi R}{3}$

(B)  $\frac{5\pi R}{3}$

(C)  $\frac{2\pi R\sqrt{3}}{3}$

(D)  $2\pi R\sqrt{3}$

(E)  $4\pi R$

18. The seven digits of the telephone number  $\overline{aaabbbb}$  add up to the two digit number  $\overline{ab}$ . What is the sum  $a+b$ ?

电话号码  $\overline{aaabbbb}$  中的七个数字相加的和是两位数  $\overline{ab}$ . 问  $a+b$  是多少？

(A) 8

(B) 9

(C) 10

(D) 11

(E) 12

19. 60 apples and 60 pears are packed into boxes so that each box contains the same number of apples, and no two boxes contain the same number of pears. What is the largest possible number of boxes that can be packed in this way?

将 60 个苹果和 60 个梨装入盒子中，使得每个盒子中包含相同数量的苹果，并且没有两个盒子中有相同数量的梨。问这样最多能装多少个盒子？

(A) 20

(B) 15

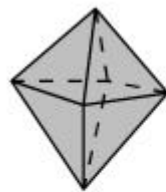
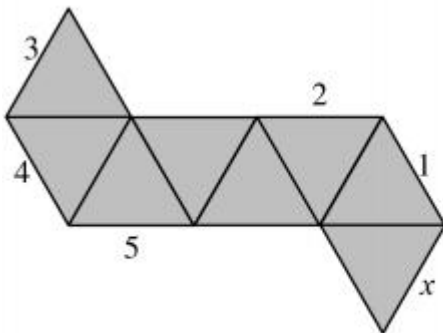
(C) 12

(D) 10

(E) 6

20 • The diagram shows a net of an octahedron. When this is folded to form the octahedron, which of the labelled line segments will coincide with the line segment marked with the x?

下面是一个八面体的平面展开图。当把它折成八面体时，用 x 标记的线段会和标记哪个数字的线段重叠在一起？



(A) 1

(B) 2

(C) 3

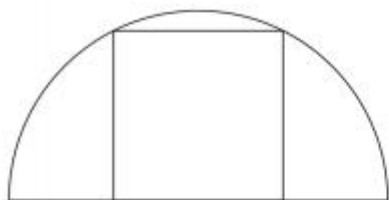
(D) 4

(E) 5

**Part 3: 10 problems, 5 points each | 第三部分: 10 道题目, 每题 5 分**

**21 •** A square has two of its vertices on a semicircle and the other two on the diameter of the semicircle as shown. The radius of the circle is 1 cm. What is the area of the square, in  $\text{cm}^2$ ?

如图所示, 正方形的两个顶点位于半圆上, 而另外两个顶点位于半圆的直径上。圆的半径是 1 厘米。问正方形的面积是多少平方厘米?



(A)  $\frac{4}{5}$

(B)  $\frac{\pi}{4}$

(C) 1

(D)  $\frac{4}{3}$

(E)  $\frac{2}{\sqrt{3}}$

**22 •** Two dots are marked in a disc that is rotating around its centre. One of them is 3 cm further than the other to the centre of the disc and moves at a constant speed that is 2.5 times as fast as the other. What is the distance from the centre of the disc to this far point, in cm?

在一个围绕其中心旋转的圆盘上标记了两个点。其中一个点距圆盘中心的距离比另一个点远 3 厘米, 并且这个点移动的恒定速度是另一个点的 2.5 倍。问较远的点到圆盘中心的距离是多少厘米?

(A) 10

(B) 9

(C) 8

(D) 6

(E) 5

**23 •** The integers from 1 to 99 are written in ascending order without gaps. The sequence of digits is then divided into triplets of digits:

从 1 到 99 的整数按照升序没有间隙的写出。然后将这个数字序列分隔成三元数字组:

123456789101112... 979899 T (123)(456)(789)(101)(112)... (979)(899)

Which of the following is not one of the triplets?

以下哪一项不在这些三元数字组中?

(A) (222)

(B) (444)

(C) (464)

(D) (646)

(E) (888)

**24 •** How many planes pass through exactly three vertices of a given cube?

有多少个平面通过给定立方体的恰好三个顶点?

(A) 1

(B) 2

(C) 4

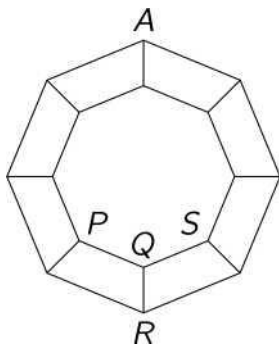
(D) 8

(E) 12

**25 •** A graph consists of 16 vertices and some edges that connect them, as in the picture. An ant is now at the vertex labelled A. At each move, it can walk from one vertex to any neighbouring vertex crawling along a connecting edge. At which of the vertices labelled F, Q, R, S, T can the ant be after 2019 moves?

如图所示, 一个图由 16 个顶点和一些连接顶点的边组成。蚂蚁现在位于标记为 A 的顶点处。在每次移动时, 它

可以从一个顶点爬到有边相连的任何相邻顶点。经过 2019 次移动，蚂蚁有可能停留在图中标记为 P, Q, R, S, T 的哪些顶点？



- (A) only P, R or S, not Q and T    只可能是 P, R 或 S, 但不可能是 Q 和 T
- (B) only P, R, S or T, not Q    只可能是 P, R, S 或 T, 但不可能是 Q
- (C) only Q    只可能是 Q
- (D) only T    只可能是 T
- (E) all of these are possible    这些顶点都可能

26. The positive integers  $a$ ,  $b$ , and  $c$  each have three digits, and for each integer the first digit is the same as its last digit. Also  $b = 2a + 1$  and  $c = 2b + 1$ . How many possibilities are there for the integers?

正整数  $a$ ,  $b$  和  $c$  都是三位数，并且对于每个整数，其首位数字和末位数字相同。另外， $b = 2a + 1$ ，以及  $c = 2b + 1$ 。问整数  $a$  的取值有多少种可能？

- (A) 0    (B) 1    (C) 2    (D) 3    (E) more than 3 多于 3

27. On each vertex of a square, one positive integer is placed. For any two numbers joined by an edge of the square, one is a multiple of the other. However, for any two diagonally opposite numbers, neither is a multiple of the other. What is the smallest possible sum of the four numbers?

在正方形的每个顶点上，放置一个正整数。对于有正方形边连接的任何两个数，一个数是另一个数的倍数。但是对于相对的角上的两个数，他们之间没有整倍数关系。问这四个数之和的最小可能值是多少？

- (A) 12    (B) 24    (C) 30    (D) 35    (E) 60

28.  $\{10, 20, 30, 40, 50, 60, 70, 80, 90\}$  What is the least number of elements we have to delete from the above set so that the product of the elements remaining in the set is a perfect square?

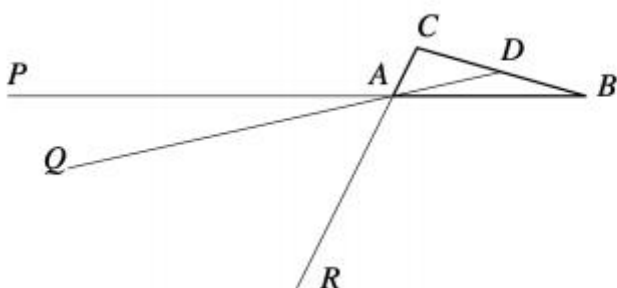
我们必须从上面的集合中去掉最少几个元素，使得剩下的所有元素的乘积是一个完全平方数？

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

29. Given triangle  $ABC$  of area  $S$ , let  $D$  be the midpoint of  $BC$ . Take points  $P, Q, R$  on lines  $AB, AD, AC$ , respectively, as shown in the picture, and such that  $AP = 2 \cdot AB$ ,  $AQ = 3 \cdot AD$  and  $AR = 4 \cdot AC$ .

What is the area of triangle  $PQR$ ?

给定面积为  $S$  的三角形  $ABC$ ，设  $D$  为  $BC$  的中点。如图所示，分别在直线  $AB, AD, AC$  上选取点  $P, Q, R$ ，满足  $AP = 2AB$ ， $AQ = 3AD$  和  $AR = 4AC$ 。问三角形  $PQR$  的面积是多少？



- (A)  $S$  (B)  $2S$  (C)  $3S$  (D)  $\frac{1}{2}S$  (E) 0 (i.e.  $P, Q, R$  are collinear 即  $P, Q, R$  共线).

30. If any digit of a given 4-digit number is deleted, the resulting 3-digit number is a divisor of the original number. How many 4-digit numbers have this property?

删除给定四位数中的任何一个数字，得到的三位数都是原数的约数。问这样的四位数有多少个？

- (A) 5 (B) 9 (C) 14 (D) 19 (E) 23

答案: DBECB BCBAB CCEBD BACDE AEBDC CDBAC